

Aggregate (cost=12866.62..12866.63 rows=1 width=64) (actual time=48.005..51.502 rows=1.00 loops=1)
 Buffers: shared hit=9602 read=60
 -> WindowAgg (cost=12832.65..12852.05 rows=971 width=30) (actual time=47.927..51.483 rows=99.00 loops=1)
 Window: w2 AS (ORDER BY (count(*)), m.song_title ROWS UNBOUNDED PRECEDING)
 Storage: Memory Maximum Storage: 17kB
 Buffers: shared hit=9602 read=60
 -> Sort (cost=12832.63..12835.06 rows=971 width=22) (actual time=47.924..51.428 rows=99.00 loops=1)
 Sort Key: (count(*)), m.song_title
 Sort Method: quicksort Memory: 29kB
 Buffers: shared hit=9602 read=60
 -> WindowAgg (cost=12765.06..12784.46 rows=971 width=22) (actual time=47.861..51.391 rows=99.00 loops=1)
 Window: w1 AS (ORDER BY (count(*)), m.song_title ROWS UNBOUNDED PRECEDING)
 Storage: Memory Maximum Storage: 17kB
 Buffers: shared hit=9602 read=60
 -> Sort (cost=12765.04..12767.46 rows=971 width=14) (actual time=47.853..51.356 rows=99.00 loops=1)
 Sort Key: (count(*)) DESC, m.song_title
 Sort Method: quicksort Memory: 28kB
 Buffers: shared hit=9602 read=60
 -> Finalize GroupAggregate (cost=12424.13..12716.86 rows=971 width=14) (actual time=47.648..51.315 rows=99.00 loops=1)
 Group Key: m.song_title
 Buffers: shared hit=9602 read=60
 -> Gather Merge (cost=12424.13..12695.50 rows=2330 width=14) (actual time=47.634..51.204 rows=297.00 loops=1)
 Workers Planned: 2
 Workers Launched: 2
 Buffers: shared hit=9602 read=60
 -> Sort (cost=11424.11..11426.54 rows=971 width=14) (actual time=42.758..42.769 rows=99.00 loops=3)
 Sort Key: m.song_title
 Sort Method: quicksort Memory: 28kB
 Buffers: shared hit=9602 read=60
 Worker 0: Sort Method: quicksort Memory: 28kB
 Worker 1: Sort Method: quicksort Memory: 28kB
 -> Partial HashAggregate (cost=11366.22..11375.93 rows=971 width=14) (actual time=42.674..42.695 rows=99.00 loops=3)
 Group Key: m.song_title
 Batches: 1 Memory Usage: 65kB
 Buffers: shared hit=9586 read=60
 Worker 0: Batches: 1 Memory Usage: 65kB
 Worker 1: Batches: 1 Memory Usage: 65kB

-> Nested Loop (cost=3170.78..11299.05 rows=13435
width=6) (actual time=15.237..40.266 rows=10567.00 loops=3)
Buffers: shared hit=9586 read=60
-> Parallel Hash Join (cost=3170.49..10930.91
rows=13435 width=4) (actual time=15.192..35.313 rows=10567.00 loops=3)
Hash Cond: (p.user_id = u.user_id)
Buffers: shared hit=8684 read=60
-> Parallel Bitmap Heap Scan on playlists p
(cost=737.45..8427.17 rows=26937 width=20) (actual time=5.124..17.403 rows=21130.00
loops=3)

Recheck Cond: (added_date >
'2026-03-01'::date)

Heap Blocks: exact=3174
Buffers: shared hit=7353 read=60
Worker 0: Heap Blocks: exact=2096
Worker 1: Heap Blocks: exact=2082

-> **Bitmap Index Scan on**
idx_playlists_added_date (cost=0.00..721.29 rows=64649 width=0) (actual
time=3.920..3.920 rows=63390.00 loops=1)

Index Cond: (added_date >
'2026-03-01'::date)

Index Searches: 1
Buffers: shared hit=1 read=60

-> Parallel Hash (cost=2066.29..2066.29
rows=29339 width=16) (actual time=9.917..9.918 rows=16656.67 loops=3)
Buckets: 65536 Batches: 1 Memory Usage:
2912kB

Buffers: shared hit=1331
-> Parallel Seq Scan on users u
(cost=0.00..2066.29 rows=29339 width=16) (actual time=0.015..5.854 rows=16656.67
loops=3)

Filter: ((subscription_type)::text =
'premium'::text)

Rows Removed by Filter: 16677
Buffers: shared hit=1331

-> Memoize (cost=0.30..0.33 rows=1 width=10) (actual
time=0.000..0.000 rows=1.00 loops=31701)

Cache Key: p.song_id
Cache Mode: logical
Hits: 13573 Misses: 100 Evictions: 0 Overflows: 0
Memory Usage: 11kB

Buffers: shared hit=902
Worker 0: Hits: 8858 Misses: 100 Evictions: 0
Overflows: 0 Memory Usage: 11kB

Worker 1: Hits: 8970 Misses: 100 Evictions: 0
Overflows: 0 Memory Usage: 11kB

-> **Index Scan using music_pkey on music m**
(cost=0.29..0.32 rows=1 width=10) (actual time=0.002..0.002 rows=1.00 loops=300)

Index Cond: (song_id = p.song_id)

Index Searches: 300

Buffers: shared hit=902

Planning:

Buffers: shared hit=22

Planning Time: 0.457 ms

Execution Time: 51.663 ms